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Q1. Personal Data & Model Performance

- a) Accuracy Change: Removing personal data (PII) reduces the Feature Space. For example, in sentiment analysis, knowing a user's location might help a model understand regional sarcasm, but removing it forces the model to rely purely on Natural Language Processing (NLP).
- b) Performance vs. Quality: This is a conflict between Empirical Accuracy and Model Robustness. A model using personal data might achieve 95% accuracy on a specific dataset but drop to 60% when tested on a different demographic because it learned "who" is talking rather than "what" is being said.
- c) Ethical Justification: Reference the Principle of Proportionality. Even if personal data increases accuracy by 2%, the risk of a data breach or unauthorized profiling is a disproportionate cost to user privacy.

Q2. Identifying & Reducing Bias

- a) Observed Bias: A famous study showed that popular perspective APIs often flagged African American Vernacular English (AAVE) as "toxic" at significantly higher rates (sometimes 2x higher) than Standard American English, even when the sentiment was positive.
- b) Source of Bias: Include Sampling Bias. If 90% of your "Real News" dataset comes from mainstream Western media, the model will naturally struggle to validate local news from other cultures, often defaulting to a "Fake" classification.
- c) Reduction Methods: * Counterfactual Fairness: Testing if the model's prediction changes if you simply swap a "male" name for a "female" name in the text.
  - Reweighting: Assigning higher importance to minority class samples during the training phase to prevent the model from ignoring them.

Q3. Explainability & Transparencya) Importance: In critical sectors like healthcare or legal tech, explainability is a safety requirement. If an AI flags a tweet as "inciting violence," a human moderator needs to see the specific linguistic triggers to ensure it wasn't just a metaphor or song lyrics.

- b) Transparency Tools: Use LIME (Local Interpretable Model-agnostic Explanations). Unlike SHAP, which looks at the whole model, LIME creates a simpler, local model around a single prediction to explain *that specific case*.
- c) Risks of "Black Box": This creates Algorithmic Injustice. If a user is banned based on a black-box model, they cannot "meaningfully appeal" because the reasoning is hidden, violating the principle of Due Process.

Q4. Data Protection & Legal Risks

- a) Compliance: Beyond India's DPDP, mention Privacy by Design. This means the system should be built to protect data by default (e.g., using Differential Privacy to add "noise" to data so individuals cannot be identified).
- b) User Rights: * Right to Opt-out: Users should be able to prevent their data from being used to train future versions of the AI.
    - Right to Portability: Users should be able to request a copy of the data the model has "learned" about them.
- c) Legal Risks: * Algorithmic Disgorgement: A legal remedy where a company is forced to delete an entire trained model because it was built using illegally obtained data (as seen in recent FTC rulings).

Q5. Public Deployment & Responsibility

- a) Ethical Issues: Filter Bubbles and Echo Chambers. If a misinformation detector is biased, it may inadvertently silence one side of a political debate while amplifying another, fundamentally damaging democratic discourse.
- b) Responsibility: Introduce the concept of "Human-in-the-loop" (HITL). Developers are responsible for the Model Audit, while the organization is responsible for the Impact Assessment.
- c) Why not Fully Automated? AI lacks Contextual Awareness. An AI might flag a historical quote about a war as "promoting violence" because it cannot distinguish between a history lesson and a call to action.